One Inverter

The loop that continuation forces, what its center reads, and compilers locked inside compilers

Driven by Dean A. Kulik ·

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Abstract

A compiler that may never stand still and must keep its way home has, at the bottom, one bit and one choice: NOT is the only map of one bit without a fixed state. Spread over a register of n bits it becomes one loop, τ: shift one place and return the departing bit complemented, the twisted ring of digital design. Read k ticks at a time, τ is complement feedback on k-bit symbols, so the two-bit rule that the ring work reached by filtering all 256 two-bit maps (reversible 24, no fixed state 9, no preferred bit 1: the complement) is the one-bit loop read two ticks at a time; at width two the loop is the Gray quarter-turn 00 → 01 → 11 → 10. A loop of one-bit cells with k inverters is conjugate to plain rotation when k is even, which holds two fixed words, and to τ when k is odd, which holds none: a latch against a clock, and continuation keeps the clock. τ^n is the complement, its orbits have lengths 2n/d for odd d dividing n (Golomb's complemented cycling register), and the words on the shorter orbits are exactly the alternating lane words $\Lambda_q(u) = u\|\bar{u}\|u\|\ldots\|u$ with q odd.

The yes/no bit at the end of the ring's inward route is the XOR of the word's two central bits, its innermost mirror lamp. Run under τ, the center reads every adjacent-bit difference once, with one known inversion: the n readings XOR to 1, and n − 1 of them with the face bit return the word. The lift Λ_q commutes with τ, with reversal, complement and the partner P, keeps the face, the canonical order and the center, and multiplies Hamming distance by q. The center trace of a word has minimal period p exactly when the word is $\Lambda_{n/p}$ of its first p bits, so a 192-bit state is certified as three locked 64-bit compilers by its center alone. Off the lock, a reference lane and q − 1 defect lanes carry the state exactly; closed feedback rotates the defect read two places per tick and never changes its weight; one revolution acts on the defects by a linear map of order exactly q; and an input pair enters as a two-bit deviation whose arrival at the center and at the reference lane is fixed by the lane count and the parity of n/2. Where an input lands in the address is the parabola $c - c^2/2N$, plus a palindrome term that peaks at $2^h/4$, plus a remainder of less than h + ½ slots. Every statement is proved, and every one is checked by one sealed notebook: 51 checks, exhaustive to widths 14 to 22 and sampled to 1,920 bits.

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How the paper runs

The paper is written from its end, as the series has been. Section 1 states the finished object: an input routed to one bit, and the loop that must run it. Section 2 fixes the ring and its own moves. Sections 3 to 8 follow the finished object back along the path that produced it, from one bit to the interface between locked compilers. Each section opens on the hole the previous result left and names what must fill it by its shape and its neighbours; no section found its result by trying candidates against a readout. Section 9 pulls back. Statements first reached in GPT, Gemini or Perplexity runs between 18 and 23 October 2026 are marked with their source, and every one was rerun here with independent code. The predictions hashed before their runs are in Appendix A; Appendix B is the notebook that reruns every statement.

1. The held end

The compiler is the constant; the input is the route through it to one bit. An n-bit word enters ring n and is carried inward, ring by ring, to the two-bit root 00 or 01, and the root's second bit is the answer, yes or no. Nothing about that answer waits for the route. It is

$$r(x) = x_{h-1} \oplus x_{n-h}, \quad h = \lfloor n/2 \rfloor,$$

the difference across the innermost mirror pair (Theorem 5.1). Every other route bit is another mirror difference or a which-way bit, so an 8-bit input fired at the compiler falls to one of 128 yes-routes or one of 128 no-routes, and a 256-bit input to one of 2^{255} of each.

This paper holds four statements about that compiler when it is required to run.

- **The loop.** A register that may never stand still and must keep its way home runs as τ, one place per tick with the departing bit complemented. Every other admissible loop of inverters is τ in disguise (Section 3).
- **The reader.** Under τ the center reads one adjacent difference per tick; over n ticks it reads all of them once, with one known inversion, and that returns the word (Section 5).
- **The lock.** q copies of an s-bit compiler laid alternately complemented around the register run as one compiler exactly when q is odd. The ring's operations see the small compiler inside the large one, and the center alone certifies it (Sections 4 and 7).
- **The interface.** Off the lock, the state is one reference compiler and q − 1 defect compilers. The defects rotate and never heal; an input is a two-bit deviation with a known time of arrival (Section 8).

The running example is the 64-bit word u = 446f6e7420666f72, the ASCII of "Dont for", the first eight bytes of the Ovaltine stream used across the threads. Table 1 collects its readings; each is produced by the notebook of Appendix B.

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Reading	Value	Where
υ (64 bits)	446f6e7420666f72	running example
ring address of υ	3b49bcdcedd4e462	Section 7; counting rank
yes/no r(υ); face	0; 0	Theorem 5.1
lamps $g_0 \dots g_{31}$, outer to inner	00001010 10011001 00001000 01110000	Theorem 5.1
center trace $y_0 \dots y_{63}$ under τ	00110000 01010101 01011000 11001011 11100110 01011000 11011001 01001110	Theorem 5.4
τ-orbit length of υ	128	Theorem 4.1
$\Lambda_3(\upsilon)$, 192 bits	446f6e7420666f72 bb90918bdf99908d 446f6e7420666f72	Theorem 4.2
address of $\Lambda_3(\upsilon)$	3b49bcdccb9d2d2b 3aa42831cb4f5581 dfa12a793ffbdf03	Section 7
center trace period of $\Lambda_3(\upsilon)$	64; 63 readings and the face return υ	Theorem 7.3

Table 1. The running example. Hexadecimal words are grouped in 64-bit lanes; bit strings in bytes.

2. The object and its moves

Fix $n \geq 2$ and write $N = 2^n$, $M = N/2$, $h = \lfloor n/2 \rfloor$ and $H = 2^h$. A word $x = x_0\, x_1 \dots x_{n-1}$ is read as an integer with x_0 most significant. J reverses a word and C complements it. The partner is $P(x) = J(x)$ when $J(x) \neq x$ and $P(x) = C(x)$ when x is a palindrome; P is an involution without fixed points. The ring R_n lists the representatives $x < P(x)$ in increasing order at addresses 0 to $M - 2$, the largest at M and its partner at $M - 1$, and places $P(R_n[i])$ at $N - 1 - i$ **[STANDS: 1]**. Its location and its inverse cost O(n) [1, 2], and its geometry folds into a unit square whose diagonal carries the palindromes [5]. $\mathrm{canon}(x) = \min(x, P(x))$ is the representative of the pair and $\mathrm{face}(x) = [x \neq \mathrm{canon}(x)]$ the side. The lamp of mirror pair j is $g_j(x) = x_j \oplus x_{n-1-j}$, $j = 0$ outermost; the lamps are the mirror differences δ_j of *Against Its Own Reverse* [3], the residue $x \oplus J(x)$ read pair by pair, and J, C and P fix every one of them. The inward route of [3, Section 4] removes the middle bit at odd steps and inverts the E/G lift at even steps, $E_m(z) = \mathrm{canon}(m\ z\ m)$ and $G_m(z) = 0\ P^m(z)\ 1$, down to the root 00 or 01 **[STANDS: 3]**.

Two pieces of notation carry the rest of the paper. rotl_k moves every bit k places toward the front, cyclically: $(\mathrm{rotl}_k\, x)_i = x_{i+k \bmod n}$. 1_k is the word whose last k bits are 1 and the others 0, the integer $2^k - 1$.

The ring has its own moves. T is the successor, $T(R_n[i]) = R_n[i + 1 \bmod N]$. GPT's run of 21 October defined $S = PC{\cdot}T^{-1}$ and $B = S^2$, and reported that T and its complement conjugate $T_C = CTC$ never agree.

Proposition 2.1 (the ring's own moves). (a) $PTP = T^{-1}$. (b) $PC = CP$, and $S^2 = T_C\, T^{-1}$. (c) $B = T_C\, T^{-1}$ has no fixed point for n = 2 to 22; equivalently the permutation matrices of T and T_C have Frobenius inner product 0. (d) $\langle T, P, C \rangle$ is the full symmetric group on the 2^n words for n = 3 to 8; at n = 2, where P = C, it is the dihedral group of order 8. **[(a), (b) proved; (c), (d) RUNS, (c) to n = 22, (d) sealed]**

Proof. (a) By the facing law, $PTP(R[i]) = PT(R[N - 1 - i]) = P(R[N - i]) = R[i - 1]$. (b) If x is not a palindrome neither is Cx, and $P(Cx) = J(Cx) = C(Jx) = C(Px)$; if x is a palindrome so is Cx, and $P(Cx) = x = C(Px)$. From (a), $T^{-1}P = PT$, so $S^2 = PC(T^{-1}P)CT^{-1} = PCPTCT^{-1} = CTCT^{-1}$. (c) and (d) are computed: (c) through the complement map on addresses, $\kappa(i) = $ address of $C(R[i])$, since B has a fixed point exactly when $\kappa(i) = \kappa(i - 1) + 1$ for some address i; (d) by 2-transitivity and an explicit transposition, which force the symmetric group by Jordan's theorem [14]. ■

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The successor is the one native move that Section 7 shows the lift does not carry. The cycle census of S (32 + 16 + 16 at n = 6, up to 936 + 28 + 28 + 16 + 16 at n = 10; GPT, rerun to n = 14 in [7, §135]) is recorded and not used here.

3. One bit, one inverter

The hole. The ring is finished, and continuation (C1: no distinction is terminal) says the compiler has to run. What runs it is fixed by its shape before any rule is tried. It acts on the register; it carries its own way home, so it is a bijection; it has no terminal state, so no fixed word; it is built from the bottom, one bit, with no constant brought in from outside; and it prefers no position and no bit.

Theorem 3.1 (one bit). On one bit the only map without a fixed state is NOT. **[proved; RUNS]**

Proof. The maps of {0, 1} are the identity, the two constants and NOT; each of the first three fixes a point. ■

Spread over a register, NOT can act only where a bit leaves. The loop

$$\tau(x_0\, x_1 \ldots x_{n-1}) = x_1 \ldots x_{n-1}\, \neg x_0, \quad \text{that is,} \quad \tau(x) = \mathrm{rotl}_1(x) \oplus 1_1,$$

shifts one place and returns the departing bit complemented. It is the twisted ring or Johnson counter of digital design, and the pure cycling register with complement feedback of shift-register theory [8]. Figure 1 shows it on the first sixteen bits of υ.

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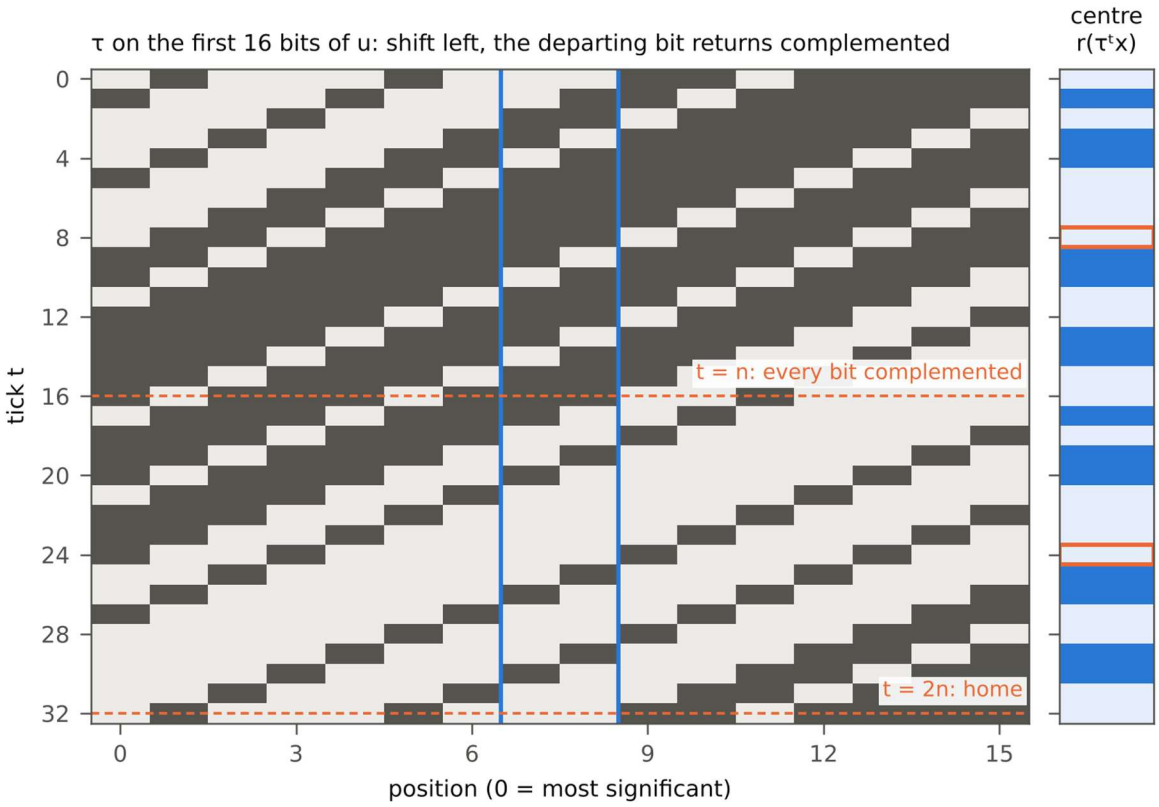


Figure 1. The loop τ on x = 446f, the first sixteen bits of u, for ticks 0 to 32. Each row is the word after t ticks, dark for 1. The blue lines enclose the central pair, and the strip at right is the center reading $r(\tau^t x)$. At t = n = 16 every bit is complemented; at t = 2n the word is home. The boxed readings are taken across the twisted boundary (t = n/2 and t = n/2 + n), where the reading is the edge difference flipped (Theorem 5.4).

Theorem 3.2 (the loop read k ticks at a time). For $0 \le k \le n$,

$$\tau^k(x) = \mathrm{rotl}_k(x) \oplus 1_k.$$

So τ^k shifts k places and returns the departing k-bit block complemented; $\tau^n = C$ and $\tau^{2n} = I$. **[proved; RUNS every word and every k, n = 2 to 16]**

Proof. By induction, $\tau^{k+1}(x) = \mathrm{rotl}_1(\mathrm{rotl}_k x \oplus 1_k) \oplus 1_1 = \mathrm{rotl}_{k+1} x \oplus 1_{k+1}$, because for $k < n$ the k ones of 1_k move one place forward without wrapping: $\mathrm{rotl}_1(1_k) = 1_{k+1} \oplus 1_1$. At $k = n$, 1_n is all ones. ■

The two-bit rule of the ring work is therefore not a separate choice. Closed two-bit feedback with rule q on the departing pair is

$$F_q(x_0\, x_1 \ldots x_{n-1}) = x_2 \ldots x_{n-1}\, q(x_0\, x_1),$$

with q the identity (return, F_{ret}), $q = C_2$ (complement, F_c) or $q = J_2$ (reverse, F_{rev}). The open port F_b puts an arbitrary incoming pair b where $q(x_0\, x_1)$ stood.

Theorem 3.3 (the two-bit filtration). (a) F_q has a fixed state iff q has a fixed symbol. (b) Of the 256 maps q of two-bit symbols, 24 are bijective; 9 of these have no fixed symbol; and exactly one of the 9 commutes with

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exchanging the two bits: $q = C_2$, which is the partner P_2 of the two-bit ring. (c) $F_c = \tau^2$. **[proved; RUNS exhaustive, register check at n = 4, 6, 8, 10]**

Proof. (a) $F_q(x) = x$ forces $x_i = x_{i+2}$ for $i \le n - 3$, so x repeats one symbol s, and then $q(s) = s$; conversely such a word is fixed. (b) $4! = 24$, and the derangements of four symbols number 9. A q that commutes with J_2 maps the fixed symbols of J_2, 00 and 11, among themselves; a derangement exchanges them and must then exchange 01 and 10. At width two the palindromes are 00 and 11, so $P_2 = C_2$. (c) Theorem 3.2 with $k = 2$. ■

C1 alone does not choose: 9 rules survive it, among them the ring's own step T_2. The 9 contain 3 involutions, so the filter "is its own inverse", applied in [7, §139] between C1 and symmetry, is not needed (Gemini, 22 October).

Proposition 3.4 (the quarter-turn). At width two, τ_2 is the cycle $00 \to 01 \to 11 \to 10 \to 00$, the Gray order of two bits. The bijections of two bits that keep one-bit adjacency, have no fixed symbol and have order four are exactly τ_2 and τ_2^{-1}, and each squares to C_2. The ring step T_2 ($00 \to 10 \to 01 \to 11$) breaks two of the four edges of the square, and T_2^2 flips the last bit. **[proved; RUNS; sealed]**

Proof. The adjacency-keeping bijections of two bits are the symmetries of the square 00–01–11–10, the dihedral group of order 8. Its elements of order four are the two quarter-turns, which fix no corner, and their square is the half-turn, which exchanges opposite corners, $00 \leftrightarrow 11$ and $01 \leftrightarrow 10$: that is C_2. τ_2 is read off its definition. ■

The ledger [7, §140] spoke of two square roots of the complement: half a step in time, the one-bit loop, and a quarter-turn in space, the Gray rotation. At width two they are one map.

Theorem 3.5 (inverter parity). Put an inverter on the input of each cell of a one-bit loop where the mask w has a 1: $f_w(x) = \mathrm{rotl}_1(x) \oplus w$. If wt(w) is even, f_w is conjugate to rotl_1; if wt(w) is odd, to τ. The conjugation is $x \mapsto x \oplus a$ for a word a computed in one pass. An even number of inverters therefore leaves exactly two fixed words, and an odd number none. **[proved; RUNS every mask at n = 6 to 10 and 64 masks at n = 12; sealed]**

Proof. $a \oplus f_w(x \oplus a) = \mathrm{rotl}_1 x \oplus (a \oplus \mathrm{rotl}_1 a \oplus w)$. The map $a \mapsto a \oplus \mathrm{rotl}_1 a$ is linear with kernel $\{0, 1_n\}$, and every word in its image has even weight; by dimension the image is all the even-weight words. So w moves to any word of the same weight parity, in particular to 0 or to 1_1, and a follows bit by bit from $a_0 = 0$. Rotation fixes 0…0 and 1…1; τ fixes nothing, because $\tau^n = C$ does not. ■

Theorem 3.5 is the ring-oscillator rule, and the split between $\Phi = 0$ and $\Phi = \pi$ in *One Bill, Four Currencies* [6]: an even loop is a latch that can hold, an odd loop a clock that must run. Continuation keeps only the clock. That closes the two-sheet question left in [7, §§131.8, 133.6], where both sheets were native and the reader D could not tell them apart; C1 can. Which physical interaction supplies the single net inversion is not derived here, and only its sign rule carries over **[OPEN]**.

4. Orbits and locks

The hole. τ has order 2n. A compiler of n bits might also run on shorter loops; which exist, and which words run them, is fixed by $\tau^n = C$ alone.

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Theorem 4.1 (the orbit law). Every τ-orbit has length $2p$ with $p = n/d$ for an odd divisor d of n. For such p, a word lies on an orbit whose length divides $2p$ iff $\tau^p(x) = C(x)$. The words on orbits of length exactly $2p$ number

$$G(p) = \Sigma_{e \text{ odd}, e \mid p} \mu(e)\, 2^{p/e},$$

and the number of orbits is $(1/2n)\, \Sigma_{d \text{ odd}, d \mid n} \phi(d)\, 2^{n/d}$. **[STANDS: Golomb [8, p. 172], OEIS A000016 [9]; proof below as checked from Gemini, 22 October; RUNS every word, n = 2 to 20, 2,097,148 words; sealed]**

Proof. The length L divides $2n$ because $\tau^{2n} = I$, and does not divide n because $\tau^n x = Cx \neq x$; so $L = 2p$ with $p \mid n$ and n/p odd. On the orbit τ^p is an involution, and $(\tau^p)^{n/p} = \tau^n = C$ with n/p odd gives $\tau^p x = Cx$; conversely $\tau^p x = Cx$ gives $\tau^{2p} x = x$. A word with $\tau^p x = Cx$ lies on an orbit of length $2p'$ with $p' \mid p$ and p/p' odd, and every such word satisfies it, so the 2^p solutions of Theorem 4.2 are the sum of $G(p')$ over those p'. Möbius inversion over the odd divisors [12] gives G, and Burnside's lemma over the cyclic group $\langle\tau\rangle$ gives the orbit count. ■

n	orbit length : number of orbits	orbits in all
6	4 : 1, 12 : 5	6
9	2 : 1, 6 : 1, 18 : 28	30
12	8 : 2, 24 : 170	172
15	2 : 1, 6 : 1, 10 : 3, 30 : 1,091	1,096
18	4 : 1, 12 : 5, 36 : 7,280	7,286
192	$128 : 2^{57}$, $384 : (2^{192} - 2^{64})/384$	—

Table 2. Orbits of τ. Powers of two have one orbit length; every odd factor of n opens a shorter loop. Under the two-bit clock $F_c = \tau^2$ each orbit of length $2p$ splits into two of length p, which is why the 192-bit two-bit periods are 64 and 192 (GPT and Gemini, 21–22 October).

Theorem 4.2 (the lock). Let $n = qs$. The words with $\tau^s(x) = C(x)$ are exactly the lane words

$$\Lambda_q(u) = u \,\|\, \bar{u} \,\|\, u \,\|\, \ldots \,\|\, u$$

when q is odd, 2^s of them; here $\Lambda_q(u)$ has q lanes of s bits and lane i is $u \oplus c_i$, with $c_i = 1_s$ for odd i and 0 for even i. When q is even there are none. **[proved; RUNS every word, qs ≤ 18]**

Proof. By Theorem 3.2, $\tau^s x = \mathrm{rotl}_s x \oplus 1_s$. In lanes $L_0, \ldots, L_{q-1}$, rotl_s moves lane $i + 1$ to lane i and lane 0 to the end, and 1_s complements the last lane, so $\tau^s x = (L_1, \ldots, L_{q-1}, C(L_0))$. Equality with $Cx = (C(L_0), \ldots, C(L_{q-1}))$ says $L_{i+1} = C(L_i)$ for $i \leq q - 2$ and $L_{q-1} = L_0$. The first makes $L_i = u \oplus c_i$ with $u = L_0$; the second asks $c_{q-1} = 0$, that is, q odd. ■

The short orbits of Theorem 4.1 are these locks: an orbit of length $2p$ at width n is $\Lambda_{n/p}$ of an orbit at width p (Theorem 7.1). At 192 bits the 2^{64} words $u\|\bar{u}\|u$ fill the orbits of length 128 (GPT and Gemini, 22 October; [7, §141]).

5. The center

The hole. The compiler runs, and its output is one bit per tick. What that bit reads, at each tick and over a revolution, is fixed by the root of the inward route and by the loop.

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Theorem 5.1 (the center is the innermost lamp). The inward route of x ends on the root word or with r = r(x) = $g_{h-1}(x) = x_{h-1} \oplus x_{n-h}$. The E/G branch bits of the route are $g_0, \ldots, g_{h-2}$ in inward order, outer to inner; with the middle bits and the face the route is a bijection of n bits with an exact return. **[STANDS: [3], Theorem 4, the route is the mirror lift; RUNS every word n = 2 to 14 and 1,800 words at 63 to 256 bits]**

Proof. Theorem 4 of [3] shows that the residual bits of the route are the mirror differences, innermost first when the word is grown; read inward they come outermost first and end on the innermost. Every inward step reverses or complements the interior, and both keep every lamp. ■

The yes/no is balanced: r is one XOR, so exactly half the words of every width route to 1. Table 3 gives the six inputs carried through the threads.

Stream (leading L bits)	L	yes/no r	face
Ovaltine	64	0	0
Singularity	64	1	1
Ovaltine	128	0	1
Singularity	128	0	0
Ovaltine	256	0	0
Singularity	256	1	1

Table 3. Six inputs routed to one bit (Perplexity and GPT, 22 October; rerun in [7, §135] and Appendix B).

Proposition 5.2 (one tick of closed feedback). For even n ≥ 6 and each of the three closed rules, $r(F_q x) = x_{h+1} \oplus x_{h+2}$. **[proved; RUNS every word, n = 6 to 14]**

Proof. F_q moves position i + 2 to position i for i ≤ n − 3, and the central positions h − 1 and h are at most n − 3. ■

In word coordinates the next answer reads two bits; in route coordinates it reads all n (GPT, 22 October). The word is the coordinate the compiler runs in, and the route is the coordinate it returns through.

Theorem 5.3 (the seam clock). Let n = 2h. For every word x and every t ≥ 1, $r(F_c^t x) \neq r(F_{ret}^t x)$ iff h is even and $t \equiv h/2 \pmod h$. At t = h/2 the three rules read $r(F_{ret}^t x) = g_0(x)$, $r(F_c^t x) = g_0(x) \oplus 1$ and $r(F_{rev}^t x) = x_{n-1} \oplus x_1$. When h is odd the three rules give the same center at every tick. **[proved; RUNS every word, n = 4 to 14, ticks 1 to 2n; GPT seal 27af4191…]**

Proof. $F_{ret} = \mathrm{rotl}_2$ and $F_c = \tau^2$, so $F_c^t x = \mathrm{rotl}_{2t} x \oplus 1_{2t}$ for 2t ≤ n. The two centers differ iff 1_{2t} covers exactly one of the positions h − 1 and h, that is iff n − 2t = h. Beyond t = h both streams repeat with period h, because $F_{ret}^h = I$, $F_c^h = \tau^n = C$ and r(Cx) = r(x). At t = h/2, rotl_h carries x_{n-1} and x_0 to the central positions, which then read g_0; under F_{rev} the first h bits have re-entered as swapped pairs and position h holds x_1. When h is odd the central pair is one two-bit symbol, and every rule keeps the XOR of a symbol. ■

The feedback rule is visible at the center only at the half-turn, when the outer shell is folded onto it (Gemini, 22 October), and only when n/2 is even.

Theorem 5.4 (the temporal reader). Let n = 2h, let $D_j = x_j \oplus x_{j+1 \bmod n}$ be the n adjacent differences around the word, and let $y_t = r(\tau^t x)$ for t = 0, …, n − 1. Then

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$$y_t = D_{(h-1+t) \bmod n} \oplus [t = h].$$

(a) $y_0 \oplus \ldots \oplus y_{n-1} = 1$. (b) $y_0, \ldots, y_{n-2}$ and the face bit determine x. (c) The readings at even t, which are the readings of the two-bit clock τ^2, read the h edges of one perfect matching of the n-cycle and have rank h; all n readings have rank n – 1. (d) Changing one bit of x changes exactly one reading of each phase. **[proved; RUNS every word, n = 2 to 16; GPT seal 28d1d7c8…, reproduced byte for byte]**

Proof. $\tau^t x = \mathrm{rot}_t x \oplus 1_t$, and the central pair of $\mathrm{rot}_t x$ is x_{h-1+t}, x_{h+t}. The mask 1_t covers exactly one of the central positions iff n – t = h. (a) The differences around a cycle sum to 0, and one reading is flipped. (b) The D_j determine x up to complement, and the face separates the two: P commutes with C, so $Cx > P(Cx) = C(Px)$ iff $x < P(x)$, and $\mathrm{face}(Cx) = 1 - \mathrm{face}(x)$. The last reading follows from (a). (c) The edges h – 1 + t for even t are every other edge of the cycle. (d) Changing x_j changes D_{j-1} and D_j, adjacent edges, one in each matching. ■

There are now two complete ways home. The E/G route records the path inward; the temporal trace records what the center sees while the compiler runs. The two-bit clock alone is blind to half the edges; reading every tick restores all but one relation, and the face supplies that one.

6. The sphere of readings

The hole. Dean, 22 October: "a sphere has no math on the surface therefore a space with no math is a surface of a sphere. nothing is not nothing." In exact form: what is the space of all partial readings of a word before any reader is chosen?

Theorem 6.1 (readings). The partial readings (S, v) of n bits, a set S of positions with bits v on S, ordered by extension, form the face poset of the boundary of the n-dimensional cross-polytope. Its realization is the sphere S^{n-1}, the join of n copies of S^0. The full words are its top cells, and two of them share a ridge exactly when they differ in one bit. Complement is the antipodal map and fixes no cell, and the quotient is the projective space RP^{n-1}. **[STANDS: [10]; RUNS GF(2) Betti numbers n = 1 to 7, the quotient n = 2 to 7]**

Proof. Send (S, v) to the simplex spanned by the vertices $(-1)^{v-i} e_i$, $i \in S$, of the cross-polytope $\mathrm{conv}\{\pm e_1, \ldots, \pm e_n\}$; these are exactly its proper faces. Complement sends every vertex to its antipode, and a free antipodal action on S^{n-1} has quotient RP^{n-1}. ■

At zero bits there is only the empty reading, the empty sphere S^{-1}. One bit gives two points, S^0: the first distinction. Each further bit sets two poles over the sphere already there, $K_{n+1} = K_n * S^0$. Nothing is not nothing: the first thing a space of readings holds is a pair, and no reading in it is privileged.

Theorem 6.2 (what the lamp reader sees). The lamps cannot tell a word from the same word with both bits of a mirror pair flipped, nor, at odd n, from the word with its middle bit flipped. The quotient of the reading sphere by these flips is S^{n-1} at even n and a cone, hence contractible, at odd n; its top cells are the 2^h lamp vectors. **[proved; RUNS n = 2 to 8]**

Proof. The reading sphere is the join of the h pair circles (for the pair {j, j'}, the vertices j0, j1, j'0, j'1 and four edges) and, at odd n, of the middle S^0. Flipping both bits of a pair is the half-turn of its circle; the quotient circle has two vertices and two edges, and the edges are lamp 0 and lamp 1. The join of h circles is S^{2h-1}. At odd n the middle S^0 folds to a point, and a join with a point is a cone. ■

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Proposition 6.3 (what keeps the sphere). A bijection of words keeps the sphere's adjacency iff it is a signed permutation of positions. J, C, τ and the three closed rules are. P is not: it tears exactly the edges that join a palindrome to a non-palindrome, $n \cdot 2^{n/2}$ of them at even n and $(n-1) \cdot 2^{(n+1)/2}$ at odd n, and carries each torn edge to distance n − 1. The ring step T tears edges as well. J, C and P fix every lamp; the feedback rules and T move lamps. **[proved; RUNS every edge, n = 3 to 14; the even count is GPT's, 22 October; the odd count sealed]**

Proof. For a palindrome x and a non-palindrome $y = x \oplus e_i$, $P(x) = Cx$ and $P(y) = Jy = x \oplus e_{n-1-i}$, at distance n − 1. Between two non-palindromes P = J, and between two palindromes (the middle flip at odd n) P = C, and both keep the edge. There are $2^{\lceil n/2 \rceil}$ palindromes, each with n (even n) or n − 1 (odd n) non-palindromic neighbours. ■

The partner keeps every lamp and tears the sphere only along the palindrome seam; the feedback rules keep the sphere and move the lamps. Closed feedback changes which positions face each other across the center without tearing anything.

7. Compilers inside compilers

The hole. Theorem 4.2 gives n-bit words that run like an s-bit compiler. Whether the ring's own operations see the small compiler inside the large one is fixed by how each operation treats lanes.

Theorem 7.1 (the lift is a compiler interface). Let q be odd, n = qs and $\Lambda = \Lambda_q$. Then

$$\tau_n\Lambda = \Lambda\tau_s, \quad J_n\Lambda = \Lambda J_s, \quad C_n\Lambda = \Lambda C_s, \quad P_n\Lambda = \Lambda P_s,$$

and therefore $\mathrm{canon}_n\Lambda = \Lambda\,\mathrm{canon}_s$ and $\mathrm{face}(\Lambda u) = \mathrm{face}(u)$. Λ is strictly increasing, $d_H(\Lambda u, \Lambda v) = q \cdot d_H(u, v)$, and for even s the center is inherited: $r_n(\Lambda u) = r_s(u)$. For even q, $\tau_n(\Lambda u) = \Lambda(\tau_s u) \oplus 1_1$ for every u. **[proved; RUNS 6,132 lifts exhaustive (s = 2 to 10, q = 3, 5, 7) and 200 at s = 64, 128; GPT, 23 October; the even-q identity sealed]**

Proof. Lane i of Λu is $u \oplus c_i$. τ_n shifts every lane one place and feeds lane i with the first bit of lane i + 1, which is $u_0 \oplus c_{i+1}$; since $c_i \oplus c_{i+1} = 1_s$, lane i becomes $\tau_s(u) \oplus c_i$. The last lane receives $\neg u_0$, which is the last bit of $\tau_s(u) \oplus c_{q-1}$ exactly when $c_{q-1} = 0$, that is when q is odd; for even q that bit is the only difference. J reverses the order of the lanes and each lane, and lane q − 1 − i carries the same staggering as lane i when q is odd, so JΛu = ΛJu. C complements every lane. Λ is injective, so Λu is a palindrome iff u is, and P follows from J and C. Λ is increasing because its first lane is u, which gives canon and face. Each lane of $\Lambda u \oplus \Lambda v$ is $u \oplus v$. For even s the central pair of Λu is the central pair of the middle lane, u or $\bar{u}$, and the XOR of two bits does not see a complement. ■

The one native move the lift does not carry is the ring step. No two lift words are adjacent on the global ring (s = 2 to 10, q = 3, 5, 7), so in that range T_n never maps a lift to a lift; the lifted representatives keep their order with wide gaps, 137 to 391 slots at s = 4, 2,083 to 7,681 at s = 6 and 32,903 to 129,013 at s = 8, all at q = 3 **[RUNS, measured; the gaps are GPT's, 23 October]**. Running, reflection and address are separate types: the lift carries the first two exactly and re-addresses the third. The address of the running example moves from

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3b49bcdcedd4e462 at 64 bits to 3b49bcdccb9d2d2b… at 192 and 320 bits, agreeing in its first 32 bits; the normalized difference, $-3.11\cdot10^{-11}$, is minus the local parabola correction of Theorem 8.9, to within 2^{-64}.

Proposition 7.2 (lamps of the lift). For $q = 2k + 1$ and even s, with $g = g(u)$ and g^R the same lamps read backwards,

$$g(\Lambda_q u) = (g \parallel g^R)^k \parallel g.$$

By Theorem 5.1 the E/G bits of the route of Λu are these lamps without the last, and its root is $g_{s/2-1}(u)$. **[proved; RUNS 4,172 lifts; GPT seal 0864e271…]**

Proof. Position j, in lane a at offset b, mirrors to lane $q - 1 - a$ at offset $s - 1 - b$. The two lanes carry the same staggering, so the lamp is $u_b \oplus u_{s-1-b}$. Reading $b = 0, \ldots, s - 1$ gives $g \parallel g^R$ in each of the k outer lanes and g in the first half of the middle lane. ■

The middle-bit qualifiers of the route are not claimed to repeat (GPT's caution, kept).

Theorem 7.3 (the center certifies the lock). Let n be even and p the minimal period of the center trace of x, which has period n. Then n/p is odd, $x = \Lambda_{n/p}(u)$ with u the first p bits of x, the τ-orbit of x has length $2p$, and $p - 1$ readings with the face bit return u, hence x. The words of trace period p number $G(p)$ of Theorem 4.1: at $n = 192$ the periods are 64 (2^{64} words) and 192 ($2^{192} - 2^{64}$); at 576 they are 64, 192 and 576; at 960 they are 64, 192, 320 and 960. **[proved; RUNS every word n = 2 to 16 and locks to 1,920 bits; GPT seal 55bc3df2…; orbit = 2 × period sealed]**

Proof. The trace of $\tau^p x$ is the trace of x shifted by p, so $\tau^p x$ and x have the same n readings, and by Theorem 5.4 they are equal or complementary. $\tau^p x = x$ is impossible: every orbit length has 2-adic valuation $1 + v_2(n)$ by Theorem 4.1, above $v_2(p)$. So $\tau^p x = Cx$, and Theorems 4.2, 4.1 and 7.1 give the lock, the orbit and the decoding, since the trace of Λu is the trace of u repeated n/p times. ■

The parity of Theorem 5.4(a) already forbids even repetition: q equal blocks XOR to 1 only if q is odd, so the odd-lane rule is visible from the center without reading a lane. The minimal period is read in linear time by the prefix function of Knuth, Morris and Pratt [11]. On a lock the compiler can run on its first lane and return the others through Λ. For the running example the trace of $\Lambda_3(u)$ has period 64 and its first 64 readings are the trace of u (Table 1); 63 of them and the face 0 return u. This is a restricted invariant family, not compression of arbitrary 192-bit words: a lock holds 64 bits of information in 192 (GPT).

8. The interface: defects and the input port

The hole. Sections 4 and 7 say what runs on a lock. Nothing yet says how a state enters a lock, leaves it, or is changed by input; GPT named this on 23 October as the rule of entry into, and interaction between, locked families. Its shape is fixed by Theorem 4.2: the distance from a lock must be carried somewhere, and in a reversible system it must be carried exactly.

Let $n = qs$. Write a state X in lanes $L_0, \ldots, L_{q-1}$, remove the staggering, $V_i = L_i \oplus c_i$, and take the first lane as reference and the others as defects relative to it:

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$$\upsilon = V_0, \quad E_i = V_i \oplus V_0 \quad (i = 1, \ldots, q-1).$$

For odd q, X is a lock iff E = 0. The defect read is $\Delta(X) = X \oplus C(\tau^s X)$, zero exactly on the locks.

Theorem 8.1 (defect closure). $X \mapsto (\upsilon, E_1, \ldots, E_{q-1})$ is a bijection. The lanes of $\Delta(X)$ are the cyclic differences $d_i = V_i \oplus V_{i+1}$, indices mod q; for even q the last carries an extra 1_s. Hence $d_0 \oplus \ldots \oplus d_{q-1} = 0$ for odd q and 1_s for even q, and $\Delta(X) = 0$ iff X is a lock. From Δ and any one lane, $V_{i+1} = V_i \oplus d_i$ returns X. For $q = 3$, $\Delta = (E_1, E_1 \oplus E_2, E_2)$. **[proved; RUNS 1,332,288 states exhaustive and the closure for q = 2 to 8; GPT seal 0a561f1d…]**

Proof. $\tau^s X = (L_1, \ldots, L_{q-1}, C(L_0))$ by Theorem 3.2, so $d_i = L_i \oplus C(L_{i+1}) = V_i \oplus V_{i+1}$ for $i \le q-2$, because $c_i \oplus c_{i+1} = 1_s$; and $d_{q-1} = L_{q-1} \oplus L_0 = V_{q-1} \oplus V_0 \oplus c_{q-1}$. ■

The closure law is the cyclic difference law of Theorem 5.4 one level up. There the cells are bits and the cycle is the word; here the cells are compilers and the cycle is the ring of lanes. An even number of lanes leaves a holonomy, 1_s, that no state can cancel.

Theorem 8.2 (transport). One tick of F_c acts on (υ, E) by

$$\upsilon' = F_c(\upsilon) \oplus \mathrm{top}_2(E_1),$$
$$E'_i = \mathrm{shift}_2(E_i) \mid \mathrm{top}_2(E_{i+1} \oplus E_1) \quad (1 \le i \le q-2),$$
$$E'_{q-1} = \mathrm{shift}_2(E_{q-1}) \mid \mathrm{top}_2(E_1),$$

where top_2 reads the first two bits of a lane and shift_2 drops them and appends 00. The defects run by themselves and reach the reference through one pair. After s/2 ticks,

$$E \mapsto M_q E = (E_2 \oplus E_1, \ldots, E_{q-1} \oplus E_1, E_1), \quad \upsilon \mapsto \bar{\upsilon} \oplus E_1,$$

and M_q has order exactly q; for $q = 3$, $M_3 = (1\,1;\, 1\,0)$ over GF(2) and $M_3^3 = I$. **[proved; RUNS exhaustive and 240 wide states, the order for q = 3 to 15; GPT seal 1126ebbd…; exact order sealed]**

Proof. Lane 0 receives the first pair of lane 1, which is $\mathrm{top}_2(\upsilon) \oplus 11 \oplus \mathrm{top}_2(E_1)$, so $\upsilon' = F_c(\upsilon) \oplus \mathrm{top}_2(E_1)$. Lane i receives the first pair of lane i + 1; removing υ' and c_i, the shifted part gives $\mathrm{shift}_2(E_i)$ and the incoming pair gives $\mathrm{top}_2(E_{i+1}) \oplus \mathrm{top}_2(E_1)$, because exactly one of c_i and c_{i+1} is 1_s. The last lane receives the complemented first pair of lane 0, which leaves $\mathrm{top}_2(E_1)$. For the revolution, $F_c^{s/2} = \tau^s$ by Theorem 3.2, which in V coordinates is $V'_i = C(V_{i+1})$ and $V'_{q-1} = C(V_0)$, a cyclic shift of the lanes with a complement; so $E'_i = V'_i \oplus V'_0 = E_{i+1} \oplus E_1$, with $E_q = E_0 = 0$. The coordinates E are V modulo one XOR common to all lanes, so M_q is the cyclic shift ρ of $GF(2)^q$ acting on the quotient by the all-ones vector. $\rho^q = I$, and for $0 < k < q$, $\rho^k e_0 \oplus e_0 = e_k \oplus e_0$ is neither 0 nor all-ones once $q \ge 3$. ■

The three-lane phase is not imposed: it is the order of M_3. A damaged lock does not average out; the relations between its defects cycle.

Theorem 8.3 (defects never heal; an input injects). $\Delta(F_c X) = \mathrm{rotl}_2 \Delta(X)$, so closed feedback conserves the weight of Δ. For the open port with incoming pair b and departing pair $a = x_0 x_1$, let $\varepsilon = b \oplus C_2(a)$, the deviation from closed feedback. Then $F_b X = F_c X \oplus \varepsilon$ and, reading words as integers,

$$\Delta(F_b X) = \mathrm{rotl}_2 \Delta(X) \oplus \varepsilon \oplus 2^s \varepsilon.$$

[proved; RUNS exhaustive and wide, 5,376 input checks; GPT seal 1126ebbd…]

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Proof. $\tau = \mathrm{rotl}_1 \oplus 1_1$ is affine, and $\Delta(X) = (I \oplus \mathrm{rotl}_s)X \oplus (1_s \oplus 1_n)$, whose linear part commutes with rotl_2. With $F_c = \mathrm{rotl}_2 \oplus 1_2$, $\Delta(F_cX) = \mathrm{rotl}_2(I \oplus \mathrm{rotl}_s)X \oplus (I \oplus \mathrm{rotl}_s)1_2 \oplus 1_s \oplus 1_n$, and that constant equals $\mathrm{rotl}_2(1_s \oplus 1_n)$. The port adds ε to F_cX, and Δ adds $(I \oplus \mathrm{rotl}_s)\varepsilon = \varepsilon \oplus 2^s\varepsilon$. ■

Corollary 8.4 (no closed entry). Under closed feedback no state enters or leaves a lock. A lock is entered or left only through the port, or by a projection onto the locks; such a projection keeps a way home only if E is kept beside it. **[proved]**

Proof. $\mathrm{wt}(\Delta)$ is invariant under F_c, and the locks are $\Delta = 0$. A map of all 2^n states onto the 2^s locks is many-to-one unless the 2^{n-s} values of E are kept. ■

This narrows the open entry problem: the rule that selects an entry is a rule on the port, or a decoder with its syndrome stored (Proposition 8.7).

Theorem 8.5 (arrival). Let the port deviate by ε at tick 1, with closed feedback before and after, and let $h = n/2$. Reading words as integers, the deviated and undeviated runs differ after t ticks by $\mathrm{rotl}_2^{t-1}(\varepsilon)$: one pair moving two places per tick from the end of the word toward the reference lane, whatever the state. The center registers it at

$$\text{h even: } t \equiv h/2 \text{ for } \varepsilon = 10,\ t \equiv h/2 + 1 \text{ for } \varepsilon = 01,\ \text{both for } \varepsilon = 11 \pmod h;$$
$$\text{h odd: } t \equiv (h+1)/2 \text{ for } \varepsilon = 01 \text{ or } 10 \pmod h,\ \text{and never for } \varepsilon = 11.$$

The reference lane first changes at $t = (q-1)s/2 + 1$, after the center whenever $q \geq 3$. **[proved; sealed; RUNS 297 pulses, s from 2 to 130, q = 3, 5, 7]**

Proof. F_c is affine with linear part rotl_2, so the difference of the two runs is $\mathrm{rotl}_2^{t-1}(\varepsilon)$ and does not depend on X. Bit 0 of ε sits at integer position $2(t-1)$ and bit 1 at $2(t-1)+1$, modulo n, and the central pair occupies positions $h-1$ and h. For even h, bit 1 reaches $h-1$ when $t-1 = h/2-1$ and bit 0 reaches h when $t-1 = h/2$. For odd h the central pair is one symbol, both bits of ε arrive together, and $\varepsilon = 11$ flips both central bits and leaves their XOR unchanged. The pattern repeats every h ticks because $\mathrm{rotl}_2^h = I$. The reference lane occupies positions $n-s$ to $n-1$ and is first reached when $2(t-1) = n-s$. ■

s	q	n	ε = 10	ε = 01	ε = 11	lane 0
64	3	192	48 + 96k	49 + 96k	both	65
64	5	320	80 + 160k	81 + 160k	both	129
64	7	448	112 + 224k	113 + 224k	both	193
62	3	186	47 + 93k	47 + 93k	never	63
62	5	310	78 + 155k	78 + 155k	never	125
62	7	434	109 + 217k	109 + 217k	never	187

Table 4. Ticks at which the center registers a one-tick deviation ε at the port, and the tick at which it first reaches the reference lane. The 64-bit rows are GPT's (23 October); GPT's sealed prediction omitted the recurrence every h ticks, which GPT's first run showed and GPT logged, keeping the original seal. §143 sealed the whole schedule before its run. The 62-bit rows are new: with lanes of width $s \equiv 2 \pmod 4$ a deviation in both input bits never reaches the center's answer, while either single bit does.

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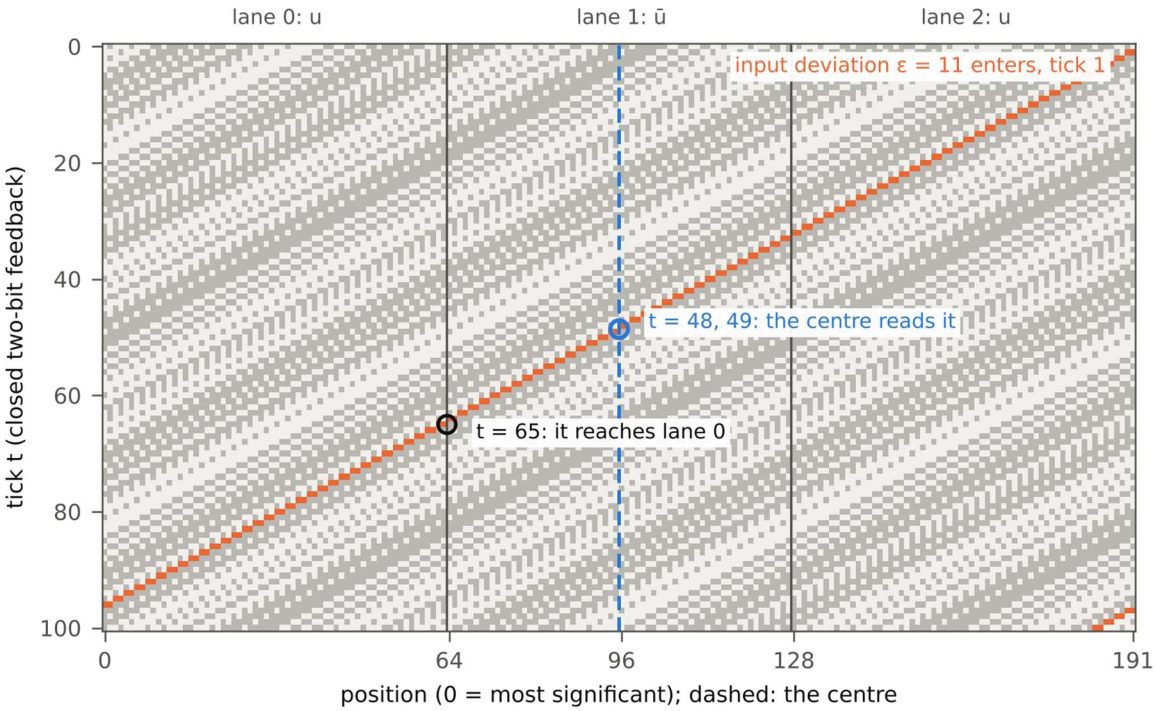


Figure 2. Three locked 64-bit compilers, u‖ū‖u with u the running example, under closed two-bit feedback for 100 ticks (grey: the state, most significant bit at left). At tick 1 the port deviates by ε = 11 (orange): one pair moving two places per tick toward the reference lane. The center (dashed) registers it at ticks 48 and 49; the reference lane first changes at tick 65; after 96 ticks the pair is back at the end of the word.

Theorem 8.6 (local center). For $q = 2k + 1$ and even s, $r_n(X) = r_s(u) \oplus r_s(E_k)$. **[proved; RUNS every state with s even, and wide; GPT seal 9d3c084d…]**

Proof. The central pair lies in lane k, which is $u \oplus c_k \oplus E_k$, and the XOR of a pair is additive and blind to a complement. ■

The center needs only the reference and the one defect lane it sits in.

Proposition 8.7 (repetition code; correct before the partner). Let s be even. Each column of a lock is a repetition: its q de-staggered bits agree. Majority over the de-staggered lanes therefore returns u through up to $\lfloor (q-1)/2 \rfloor$ wrong bits per column. If u is a palindrome, so is Λu, and a single wrong bit, $Y = \Lambda u \oplus e_i$, switches P from complement to reversal: $P(Y)$ differs from $P(\Lambda u)$ in $n - 1$ places, and majority after P returns u instead of $P(u) = \bar{u}$. Majority before P returns $\bar{u}$. Off the palindrome seam, P carries a one-bit fault to a one-bit fault. **[proved; RUNS 1,784 single faults with 204 branch switches, all double faults at q = 5; GPT seal 338b3163…]**

Proof. The first statement is the repetition code [13]. For a palindromic u, $P(\Lambda u) = C(\Lambda u)$; Y is not a palindrome, because $n = qs$ is even and a single flip of an even-length palindrome breaks it, so $P(Y) = J(Y) = \Lambda u \oplus e_{n-1-i}$, whose majority is u. Correcting first gives Λu, whose partner is $\Lambda(Pu)$. For a non-palindromic u, $J(\Lambda u) \oplus \Lambda u$ has weight $q \cdot \mathrm{wt}(Ju \oplus u) \geq 2q$, so no one-bit change of Λu is a palindrome and $P(Y) = J(\Lambda u) \oplus e_{n-1-i}$. ■

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The correction is a chosen projection, not a step of the loop, and its way home is the syndrome E (Corollary 8.4). Resolve local consistency first, then apply the conditional partner (GPT).

Proposition 8.8 (the input gate). For $y = F_b x$: (a) $J(y) = \mathrm{rev}_2(b)\cdot 2^{n-2} + \lfloor J(x)/4 \rfloor$, so the reflected rail shifts the other way, with the input pair entering reversed at the top; (b) the face of y is decided by its two outer shells unless they tie: if $\mathrm{top}_2(y) \neq \mathrm{rev}_2(b)$, then $y < J(y)$ iff $\mathrm{top}_2(y) < \mathrm{rev}_2(b)$; (c) x returns from y and the departing pair a. **[proved; RUNS every word and pair, n = 4 to 12]**

Proof. (a) and (c) are the bookkeeping of the shift. (b) $\mathrm{top}_2(y) = (y_0, y_1)$ and $\mathrm{rev}_2(b) = (y_{n-1}, y_{n-2})$; comparing them as two-bit numbers is the comparison of y with J(y) on mirror pairs 0 and 1, which decides unless both pairs agree [4, Section 3.2]. ■

On the two text streams, 216 two-bit shifts of 64-bit reads, the gate resolves 147 shifts and ties 69. The ties carry the outer pairs 01, 10, 00 and 11, only 7 of them 11, and 16 of them move the native address backwards: a tie forces neither the 11 seam nor a forward jump (GPT, correcting Gemini). The dual priority encoder of Gemini's Panel C, which marks the first and last lit lamps, reads 1.495 and 29.590 on the Ovaltine stream and 1.354 and 29.752 on the Singularity stream, spans of 29.095 and 29.398. A uniformly chosen nonzero mask of 32 lamps gives $\Sigma_k\ k\cdot 2^{31-k}/(2^{32}-1) = 0.999999993$ for the first lit lamp, 30.000000007 for the last and 30.000000015 for the span. The near-full span is the baseline of the reader, not a sign of thirty shells computing (GPT; GPT's notebook did not print the stream values, and they are reproduced here, sealed).

Theorem 8.9 (where an input lands). Let n be even and $c = A\cdot H + B$ a canonical word, with $\pi(c)$ the number of palindromes below c and $\pi_c(c) = \min(\pi(c), H/2)$ the canonical ones among them. Then

$$\mathrm{rank}(c) = c - c^2/(2N) + (\pi_c(c) - \pi(c)/2) + e, \quad \text{with } -h - \tfrac{1}{2} < e < h/2 + \tfrac{1}{2}.$$

The palindrome term lies between 0 and H/4, so the normalized address of a canonical word sits above the parabola $c/N - c^2/(2N^2)$ by at most $2^{-(h+2)} + (h+1)/(2N)$, and below it by at most $(h+1)/N$. On the parabola, the half-ring jump from $c < N/2$ to $c + N/2$ is $3/8 - c/(2N)$; natively it adds $2^{-h}(1/4 - A/H)$, to within $(3h/2 + 2)/N$. **[proved; RUNS every canonical word at n = 4 to 16 and every stream word at 64 bits; sealed]**

Proof. Theorem 1 of [3] gives $\mathrm{rank}(c) = \tfrac{1}{2}(c - \pi + \Delta) + \pi_c$, and its Theorem 7.3 gives $\Delta(c) = A(H - A) + B - 2\mathcal{R}(B, A) - \iota$ with $\iota \in \{0, 1\}$. Expanding $c - c^2/(2N)$ with $N = H^2$ leaves $e = (AB/H - \mathcal{R}(B, A)) + B^2/(2N) - \iota/2$. The rectangle count $\mathcal{R}(B, A)$ is a sum over the set bits j of B of $\lceil (A - \rho_j)/2^{j+1} \rceil$ with $0 \le \rho_j < 2^j$, and each term lies within $(-\tfrac{1}{2}, 1)$ of $A/2^{j+1}$; those $A/2^{j+1}$ sum to AB/H, and there are at most h of them. The palindrome term is $\pi/2$ while $\pi \le H/2$ and $H/2 - \pi/2$ after, at most H/4. For the jump, the term is $\pi(c)/2$ with $\pi(c) = A$ or $A + 1$ at c, and $H/2 - \pi(c + N/2)/2$ with $\pi(c + N/2) = A + H/2$ or one more at $c + N/2$. ■

Measured, e sits well inside the proved band: from $[-0.72, 0]$ at n = 4 to $[-2.72, 0]$ at n = 16, falling by about a third of a slot per added mirror pair (Figure 3). The Singularity counterfactual of Gemini's figure (x = 7468697320697320, b = 00 against 01) has native jump 0.372652843511764 and parabola jump 0.372652843454650; their difference times 2^{32} is 0.245305687, against $1/4 - A/2^{32} = 0.245305687$ predicted with A = 0133a581, and the remaining term is half a slot. The 3/8 is the parabola's coefficient for a half-ring jump, not the jump (GPT, correcting Gemini), and the discrete correction is the palindrome term.

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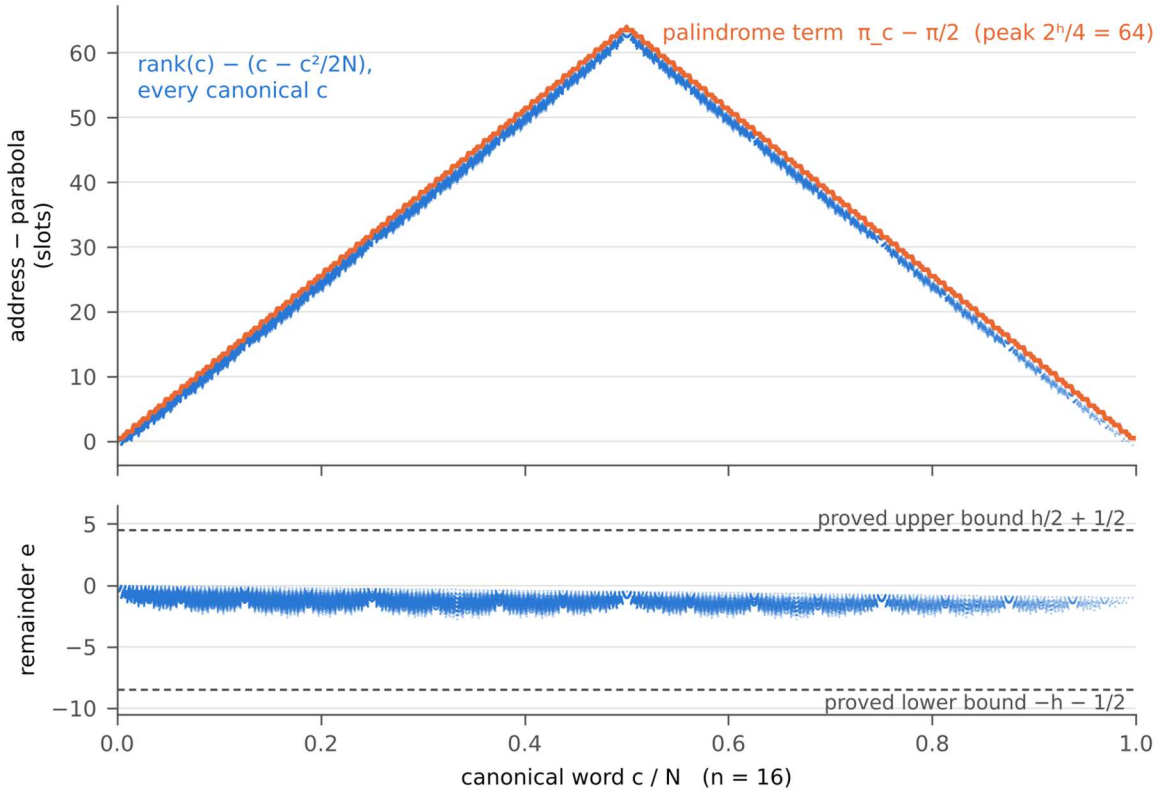


Figure 3. Where a word lands in the address, n = 16. Top: rank(c) − (c − c²/2N) for every canonical word (blue) against the palindrome term $\pi_c - \pi/2$ (orange), which peaks at $2^h/4 = 64$ slots at c = N/2. Bottom: the remainder e, between −2.72 and 0, inside the proved band (dashed).

9. Pulling back

Hold the sections at once and the paper is one statement read at three scales. Around a cycle of cells joined by edges that copy or complement, a stationary assignment exists exactly when the complementing edges are even in number: walk once around, and the value comes home unchanged iff it was flipped an even number of times.

- **Bits around the register (Section 3).** One inverter: no stationary word, and the loop must run. Even counts are latches, and continuation keeps the odd sheet.
- **Edges around the word (Section 5).** The n adjacent differences sum to zero, and the twisted boundary adds one flip: the center's n readings XOR to 1.
- **Lanes around the ring of lanes (Sections 4, 7 and 8).** A lock makes each lane the complement of the next and the wrap a copy: q − 1 complementing edges, even iff q is odd. For even q the defect lanes carry the holonomy 1_S and no lock forms.

Three further readings follow.

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The two square roots of the complement are one. Half a tick of time is the one-bit loop, and a quarter-turn of space is the Gray rotation of two bits; at width two they are the same map, and the two-bit rule the ring work reached by filtering is that map read twice.

The center is where the compiler's question sits. It is the innermost lamp before the compiler runs, the reader of one adjacent difference per tick while it runs, and the first place a deviation at the port is seen, half a revolution after it enters. The feedback rule itself shows at the center only at the half-turn, when the outer shell is folded onto it.

Running, reflection and address are separate types. The lift carries τ, J, C and P exactly and keeps the canonical order, so a small compiler runs inside a large one without the large one being built; it does not carry the ring step, and the global address resolves the small family at another scale.

10. Ledger

Statement	Status	Scope
Ring, partner, facing law; route = mirror lift; closed address	STANDS	[1], [3]
2.1 (a, b): PTP = T^{-1}; PC = CP; $S^2 = T_C T^{-1}$	proved; RUNS	every word, n = 2–22
2.1 (c): B has no fixed point	RUNS; sealed; every width OPEN	every word, n = 2–22
2.1 (d): ⟨T, P, C⟩ is the symmetric group	RUNS; sealed	n = 3–8; order 8 at n = 2
3.1 one bit: NOT	proved; RUNS	exhaustive
3.2 $\tau^k = \mathrm{rotl}_k \oplus 1_k$; $\tau^n = C$	proved; RUNS	every word, n = 2–16
3.3 filtration 256 ⟶ 24 ⟶ 9 ⟶ 1 = $C_2 = P_2$	proved; RUNS	exhaustive
3.4 τ_2 is the Gray quarter-turn	proved; RUNS; sealed	exhaustive
3.5 inverter parity: latch or clock	proved; RUNS; sealed	every mask, n = 6–10
4.1 orbit law and census	STANDS [8]; RUNS; sealed	every word, n = 2–20
4.2 the lock exists iff q is odd	proved; RUNS	every word, qs ≤ 18
5.1 center = innermost lamp; route = lamps + which-way	STANDS [3]; RUNS	n = 2–14; 1,800 wide
5.2 next read = $x_{h+1} \oplus x_{h+2}$	proved; RUNS	n = 6–14
5.3 seam clock; half-turn reads g_0	proved; RUNS	every word, n = 4–14
5.4 temporal reader	proved; RUNS	every word, n = 2–16
6.1 readings = S^{n-1}; complement antipodal; RP^{n-1}	STANDS [10]; RUNS	n = 1–7
6.2 lamp view: sphere or cone	proved; RUNS	n = 2–8
6.3 adjacency; the partner's torn edges	proved; RUNS; sealed (odd n)	every edge, n = 3–14
7.1 Λ intertwines τ, J, C, P; center; Hamming × q	proved; RUNS; sealed (even q)	6,132 lifts + 200 wide
Ring step not carried; order kept with wide gaps	RUNS (measured)	s = 2–10, q = 3, 5, 7
7.2 lamps of the lift	proved; RUNS	4,172 lifts
7.3 the center certifies the lock; period census	proved; RUNS; sealed	n = 2–16; to 1,920 bits
8.1 defect closure	proved; RUNS	1,332,288 states

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Statement	Status	Scope
8.2 transport; M_q of order q	proved; RUNS; sealed (order)	exhaustive; q = 3–15
8.3 Δ rotates; the port injects	proved; RUNS	5,376 inputs
8.4 no closed entry	proved	—
8.5 arrival, both parities of h	proved; RUNS; sealed	297 pulses
8.6 local center	proved; RUNS	exhaustive + wide
8.7 repetition code; correct before the partner	proved; RUNS	1,784 faults
8.8 input gate; ties; Panel C baseline	proved; RUNS; sealed (Panel C)	n = 4–12; 216 shifts
8.9 address = parabola + palindrome term + e	proved; RUNS; sealed	n = 4–16; streams
Hardware readings: radix-4 tagged shift, bit-reversal unit, CLZ/CTZ comparator, differential pair	PULL	Gemini
The physical interaction that supplies the net inversion	OPEN	—
Entry into and switching between lock families	OPEN (narrowed by 8.4)	—
B without fixed points at every width	OPEN	runs to n = 22
S-cycle templates; the 27 → 28 branch relation; a rung phase rule	OPEN	—
Benefit of certified locks to a learning system	OPEN	—

Table 5. Ledger. "Sealed" marks the statements whose prediction was hashed in § 143 before its first run.

Corrections logged on the way

The center's question is set by more than T. [7, §135] implied that the ring step is the only native move that sets the next question; closed feedback does too (§137).

Not the only closed surface. §136 called the sphere the only closed surface with no special point; RP^2 has none either. Orientability selects the sphere among surfaces and simple connectedness in general; RP^{63} is orientable (GPT).

Blinder than RP^{n-1}. §137 said the lamp reader sees RP^{n-1}; it sees S^{n-1} at even n and a cone at odd n (GPT, accepted in §138).

Own-inverse is redundant. §139's filter between C1 and symmetry is not needed (Gemini, §140).

One map, not two. §140 spoke of two square roots of the complement; at width two they are the same map (Proposition 3.4).

C1 alone does not choose. Gemini's first reading drew the complement from C1; nine rules survive C1 (§139).

The clock is the lane width's. Gemini's 16-tick center/seam alternation comes from 64-bit lanes and the two-bit stride for every lane count; three compilers are special because the center starts inside a lane (§139).

Not 64-specific; not a physical law. The lock holds for every lane width; the loop law Φ = π carries over only as the sign rule and does not say which physical interaction realizes the twist (§141).

The pulse recurs. GPT's sealed pulse prediction omitted the second reading every h ticks. GPT's first run showed it, and GPT added it to the expectation and logged the correction, keeping the original seal. The full schedule, both parities of h, was sealed in §143 before it ran (Theorem 8.5).

The last lamp is the root. GPT's first comparison of the lifted route counted h – 1 branch bits against h lamps.

Ties, a coefficient, a baseline. In Gemini's three-panel figure a tie was read as forcing the 11 seam and a forward jump, 3/8 as the jump, and the wide Panel C span as computation. All three were corrected by GPT and are rerun here (Proposition 8.8, Theorem 8.9).

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What this paper does not claim

Everything here is finite combinatorics of binary words and what their readers return. The single net inversion is the sign rule of a ring of inverters; no physical interaction is derived, and $\Phi = 0$ and $\Phi = \pi$ are not physics here. The sphere of readings is a combinatorial sphere. Locks are invariant sub-families, not compression of arbitrary words: a lock holds s bits of information in n, and an arbitrary n-bit word holds n. No benefit to a learning system is measured. The hardware readings Gemini drew, the radix-4 shift with tagged return, the bit-reversal unit, the CLZ/CTZ comparator and the differential pair, are analogies and are marked PULL. Nothing here bears on the preimage or collision resistance of any hash function.

The next stroke

Five routes are open, each with its missing link named.

1. **Entry and switching.** Corollary 8.4 shows that closed feedback never enters a lock. The missing link is the admissible rule on the port, or the decoder with its stored syndrome, that moves a state into, out of, or between lock families. GPT's proposed test is concrete: feed sequential data through the port, keep the defect route visible, and ask whether the center's early reading (Theorem 8.5) carries information the reference lane has not yet received.
2. **The physical inversion.** Theorem 3.5 fixes the sign rule; the interaction that supplies one net inversion in a physical loop is not in hand.
3. **B at every width.** Proposition 2.1(c) runs to $n = 22$. A proof needs the complement map on addresses, κ, never to send two consecutive addresses to two consecutive addresses in the same direction.
4. **The S-cycles.** The 16- and 28-state branch templates of GPT's 21 October run, the $27 \to 28$ relation between $n = 8$ and $n = 10$, and a rung phase rule across widths.
5. **The constant in e.** The remainder of Theorem 8.9 measures $-(h/3 + 0.06) < e \le 0$ for $n \le 16$, against the proved $(-h - \tfrac{1}{2}, h/2 + \tfrac{1}{2})$. $\mathcal{R}(B, A)$ counts the bit-reversal (van der Corput) point set in a rectangle, and that set's discrepancy is the natural owner of the constant.

Conclusion

One bit with no terminal state has one map, NOT. A register built from it has one loop, τ, and every loop of inverters is either τ or a latch. The loop's center is the ring's yes/no, the innermost mirror lamp, and while the loop runs the center reads every adjacent difference once, with one inversion, which is enough to return the word. Copies of a compiler laid alternately complemented around the loop run as one compiler exactly when their number is odd; the ring's reflections see the small compiler inside the large one, and the center certifies it by its period. Off the lock the state is a reference and its defects: the defects rotate and never heal, one revolution cycles them with order q, and an input is a deviation of two bits whose arrival is fixed in advance. Where it lands in the address is a parabola, a palindrome tent and a few slots. None of it was put in. It came out of holding the end and reading each hole for what had to be there.

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Acknowledgment. Computation and drafting were done with Claude (Anthropic) as a working partner, which also led the audit of the other runs. Several results were first reached in independent AI-assisted runs between 18 and 23 October and are recorded with their seals in the ledger [7, §§135–143]: ChatGPT (the traversal algebra and S-cycles, closed two-bit feedback and its 192-bit periods, the seam clock, the temporal reader and the period compiler, the odd-lane partner bridge, defect transport, arrival, the local center, the fault-seam ordering and the chart-gate corrections); Gemini (the orbit-law proof, the two routes to the complement, the half-turn reading of the outer lamp, the three-panel plot and the hardware readings); Perplexity (the ring, route and address rebuilds of 18 to 20 October). Every number in this paper was recomputed by the notebook of Appendix B.

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Appendix A. Sealed predictions

Each prediction was written with the reading it predicts and the readout it returns through, and its text was hashed before the code that tests it ran. Each runner prints the digest before it computes. The digests of the GPT runs are GPT's own, printed by GPT's notebooks and reproduced byte for byte here.

Prediction	SHA-256	Result
§135 ring core, center = central XOR	`9fec65cd1425b1949dffbf80776356d1 65a822ac6c589776f9475b1bcce5934c`	RUNS
§136 the sphere of readings	`8128a58900769a6fd618c3e54754597c 47cf00cfbd8ccfd1989bef502c18269c`	RUNS; one reading corrected
§137 complement quotient; corners	`dc5b937a154ee804b504e1b7855427d7 406e81fc152c70a217e0c3cfc1f0c24f`	RUNS; reader claim corrected
§138 lamp view; seam clock	`bf083397c8aab6eae5a94e88d9dbaa97 a47956171b4d03e9e9a7ba3f0e1c7ff4`	RUNS
§139 two-bit filtration	`1110777a10b2e6e07723391698b470b7 6eb424f9455022515110ba308c9f0cdd`	RUNS
§140 one-bit root	`99c261a515c0be32864588c79617d318 81c3efa352a68fb239af38aed621cef0`	RUNS
§141 orbit counts; lock; inverter parity	`181be0244372488df109a37c12b16369 814b74d86e326e35e1710185ac4d300b`	RUNS
GPT center seam	`27af41913aee32fd14a91658f59f60fa 54cd12203063b9a1aa8844172ac8ad24`	RUNS
GPT lane phase	`8b4f4952716dd09ff0a9a8dc66954ec3 3f627584950968184db97078ccd2d589`	RUNS
GPT defect transport	`1126ebbdbed703be71a30166e2343e04 3eb293ea4a50a09452e0f580e1b38f9d`	RUNS; recurrence added after the first run, logged
GPT cyclic closure	`0a561f1d2548938d24b00f0fed3a6d87 5e228086ca151024bcc97bf3c29e67c3`	RUNS
GPT fault seam	`338b31633481f8bd833c27d5cc612da1 05710007373a87da574d7891aa3207f1`	RUNS
GPT local center	`9d3c084db4ab853a39d6a4092dd16f42 488f2cb3d908d5cc4a438e9f27063515`	RUNS
GPT temporal center	`28d1d7c8b45becdd5d544d6db854c326 f51210c471b46d50cc414b17cd7765b6`	RUNS
GPT period compiler	`55bc3df22e64b617997ad13a07f85a12 52ca658e40884c3dae443aa78d5d3afe`	RUNS
GPT lamps of the lift	`0864e271bcc7b2dfc7cda0965a2da516 d40075a626532587ebc2770060c3b30e`	RUNS
GPT chart gate	`f93bb0757e5adc0456f3bb89a32ae8d5 d8ac4e9e12df9ff37beac25a17bbd802`	RUNS
§143 this paper (N1–N12 and the reruns)	`009815944fd8951cf95ca6bd56728f54 b5e20634dd03e2d8cc67e005019d1dea`	RUNS, all twelve new predictions

Table 6. Seals. Digests are SHA-256 of the literal prediction text.

The twelve new predictions of §143 were: N1 inverter parity for every mask (Theorem 3.5); N2 the quarter-turn (Proposition 3.4); N3 the orbit census to n = 20 (Theorem 4.1); N4 orbit length = 2 × trace period (Theorem 7.3); N5 arrival for both parities of h (Theorem 8.5); N6 the partner's torn edges at odd n (Proposition 6.3); N7 the exact order of M_q (Theorem 8.2); N8 the even-q defect of the lift (Theorem 7.1); N9 the address parabola and

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the Singularity jump (Theorem 8.9); N10 B without fixed points to n = 22 (Proposition 2.1); N11 ⟨T, P, C⟩ = S_N (Proposition 2.1); N12 GPT's unprinted Panel C values (Proposition 8.8). The full text follows as hashed.

```
NEXUS ledger section 143 - sealed predictions for the verification notebook of the paper on the forced loop.
Dean A. Kulik, QuHarmonics Research Group, ORCID 0009-0003-3128-8828. 2026-10-23. Claude leads; GPT and Gemini results
are input.
Conventions. x is an n-bit word; string index 0 is the most significant bit. J reverses, C complements, P(x)=J(x) unless
x is a
palindrome, then C(x). canon(x)=min(x,P(x)); face(x)=[x != canon(x)]. Lamps g_j = x[j] xor x[n-1-j], j=0 outermost; the
center r(x)
is the innermost lamp. tau(x)=rotl1(x) xor 1 (the departing bit returns complemented). F_c, F_ret, F_rev: closed two-bit
feedback
returning the departing pair complemented, unchanged, swapped. F_b: open port, incoming pair b. Lambda_q(u)=u|Cu|u|...|u,
q lanes of
s bits. V_i = L_i xor (i mod 2)*ones; E_i = V_i xor V_0; Delta(X) = X xor C(tau^s X). m = n/2 for even n.
NEW PREDICTIONS (not run anywhere before this seal):
N1 Inverter parity: for every mask w at n=6..10 (and 64 seeded masks at n=12), f_w(x)=rotl1(x) xor w is conjugate by an
XOR translation
   to rotl1 when wt(w) is even and to tau when wt(w) is odd; fixed states 2 and 0; orbit-length multisets equal those of
rotl1 and tau.
N2 At width 2, tau_2 is the Gray quarter-turn 00>01>11>10>00. The adjacency-keeping, fixed-point-free, order-4 two-bit
rules are exactly
   tau_2 and tau_2^-1, each squaring to C_2. The ring step T_2 is neither; T_2^2 flips one bit.
N3 Orbit law, exhaustive n=2..20: tau-orbit lengths are {2n/d : d odd, d | n}; the words of orbit length 2p number sum
over odd e | p of
   mu(e) 2^(p/e); the orbit total is (1/2n) sum over odd d | n of phi(d) 2^(n/d) (Golomb 1967; OEIS A000016).
N4 Every word at even n=2..16: tau-orbit length = 2 x the minimal period p of the center trace, n/p is odd, and x =
Lambda_(n/p)(top p bits).
N5 Pulse arrival on a lock (q odd, s even, m = qs/2), deviation eps = b xor (a xor 3) injected at tick 1, then closed
feedback:
   m even: eps=2 is seen at the center at ticks m/2 + k m, eps=1 at m/2 + 1 + k m, eps=3 at both; m odd: eps=1 and eps=2
at (m+1)/2 + k m,
   eps=3 never. The reference lane (lane 0) first changes at tick (q-1)s/2 + 1 for every eps. s in
{2,4,6,8,10,62,64,66,126,128,130},
   q in {3,5,7}, ticks 1..2n, three seeded lock words each.
N6 P tears one-bit adjacency on exactly n 2^(n/2) edges at even n and (n-1) 2^((n+1)/2) edges at odd n; every torn edge
lands at
   Hamming distance n-1; J, C, tau, F_ret, F_c, F_rev keep every edge; n=3..14.
N7 The one-revolution defect map M_q(E_1..E_(q-1)) = (E_2^E_1, ..., E_(q-1)^E_1, E_1) has order exactly q, q =
3,5,...,15.
N8 For even q: tau(Lambda_q u) = Lambda_q(tau u) xor 1 for every u (s=2..8, q=2,4,6).
N9 Address parabola, even n: for canonical c = A 2^h + B, rank(c) = c - c^2/(2N) + (pi_c(c) - pi(c)/2) + e with -h - 1/2
< e < h/2 + 1/2;
   all canonical c at n=4..16 and every stream word at n=64. Hence 0 <= pi_c - pi/2 <= 2^h/4 and the normalized parabola
error is below
   2^-(h+2) + (h+1)/N. Singularity counterfactual (b=00 vs 01 after x0=7468697320697320): (native jump - parabola jump) x
2^32
   = 1/4 - A0/2^32 within 2^-24, A0 = top 32 bits of the b=00 canonical word.
N10 PTP = T^-1 at every n (facing law); B = S^2 = T_C T^-1 with S = PC T^-1; B has no fixed point, n = 2..22.
N11 <T, P, C> is the full symmetric group on 2^n words for n = 3..8 (2-transitive and contains a transposition); at n = 2
it has order 8.
N12 GPT's unprinted Panel C means reproduce to three decimals: last lit shell 29.590 (Ovaltine) and 29.752 (Singularity),
span 29.095 and
   29.398, over 64-bit reads at two-bit stride.
REPRODUCED (run before by GPT or in sections 135-142, re-run here with independent code): tau^k is k-bit complement
feedback and tau^n = C;
one bit leaves only NOT; register C1 iff no fixed symbol; filtration 256 > 24 > 9 > 1 = C_2 = P_2 (own-inverse 3); lock
iff q odd;
route root = center = innermost lamp and E/G bits = lamps (Against Its Own Reverse, Theorem 4); six-input table; next
center read = bits
m+1, m+2; seam clock and half-turn outer lamp; temporal trace law, parity 1, decode from n-1 readings and the face, ranks
m and n-1;
reading complex S^(n-1), quotient RP^(n-1), lamp view sphere (even n) or cone (odd n); Lambda intertwines tau, J, C, P,
canon, face,
center; Hamming x q; canonical order kept with gaps (137,391), (2083,7681), (32903,129013); lamps of the lift; period
spectra at 192,
576, 960; defect coordinates, cyclic closure, transport, Delta rotation and injection, local center, majority and seam
ordering;
counting address = built ring, Ovaltine 3b49bcdcedd4e462 at 64 bits and GPT's 192- and 320-bit values; stream gate counts
147/69.
WAY HOME: the notebook prints this seal's SHA-256 before it computes and stops if it differs from the recorded digest;
every check
prints PASS with its range; each identity carries its own inverse (tau^-1; feedback inverted from the inserted pair;
Lambda^-1 = lane 0;
X <-> (u, E); the temporal decode from n-1 readings and the face bit; the address inverted by the built ring).
```

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Appendix B. The verification notebook

NEXUS_143_one_inverter_verification.ipynb is one self-contained notebook: a seal cell that stops if the text above does not hash to its recorded digest, one library cell, one cell per section of this paper, a figure cell and a ledger cell that stops unless every check passed. It uses the Python standard library, NumPy for the widest ring checks, and matplotlib only for the figures. Run in a fresh kernel it takes about 80 seconds. Its ledger, unedited:

```
2.1   PASS  facing law -> PTP = T^-1; PC = CP; S^2 = T_C T^-1  [every word, n = 2..22 (S^2 to 14)]
2.2   PASS  B = T_C T^-1 has no fixed point (<T, T_C>_F = 0)  [every word, n = 2..22]
2.3   PASS  <T, P, C> = full symmetric group S_(2^n) (2-transitive + transposition, Jordan); order 8 at n = 2  [n = 2..8]
3.1   PASS  one bit: the only fixed-point-free map is NOT  [all 4 maps]
3.2   PASS  tau^k = rotl_k xor (2^k-1) = k-bit complement feedback; tau^n = C; tau^2n = I  [every word, every k, n =
2..16]
3.3a  PASS  F_q has a fixed state iff q has a fixed symbol  [all 256 rules, n = 4, 6, 8, 10]
3.3b  PASS  filtration 256 > 24 > 9 > 1, survivor = complement = P_2 (own-inverse: 3, not needed)  [exhaustive]
3.4   PASS  tau_2 = Gray quarter-turn 00>01>11>10; quarter-turns = {tau_2, tau_2^-1}, squares = C_2; T_2 neither, T_2^2
flips one bit  [exhaustive]
3.5   PASS  inverter mask w: conjugate to rotation (wt even, 2 fixed states) or to tau (wt odd, none)  [2112 masks, n =
6..12]
4.1   PASS  orbit lengths {2n/d}, words per length by Mobius inversion, orbit totals = A000016  [every word, n = 2..20
(2,097,148 words)]
4.2   PASS  tau^s x = C x  <=>  x = Lambda_q(u), q odd; no solution for q even  [every word, q*s <= 18]
5.1a  PASS  route root = center (innermost lamp)  [every word n = 2..14; 1,800 words at 63..256 bits]
5.1b  PASS  E/G branch bits + root = the lamps, outer to inner  [same range]
5.1c  PASS  route = root + (n-2) bits + face = n bits, a bijection, exact return  [same range]
5.1d  PASS  six-input table (yes/no, face)  [Ovaltine, Singularity at 64/128/256 bits]
5.2   PASS  next center under any closed rule reads string bits h+1, h+2  [every word, n = 6..14, three rules]
5.3a  PASS  seam clock: comp/return centers differ exactly at t = h/2 mod h (h even), never (h odd)  [every word, n =
4..14, ticks 1..2n]
5.3b  PASS  at t = h/2 the center reads g0 / g0^1 / x[n-1]^x[1]  [every word, n = 4, 8, 12]
5.4a  PASS  y_t = D_(h-1+t) xor [t = h]  [every word, n = 2..16 even]
5.4b  PASS  XOR of the n center readings = 1  [same]
5.4c  PASS  n-1 readings + face return the word  [same]
5.4d  PASS  one phase rank h, both phases rank n-1; a one-bit flip moves one reading in each phase  [every word, n =
2..12 (flips n <= 10)]
6.1a  PASS  partial readings = boundary of the cross-polytope = S^(n-1) (GF(2) Betti)  [n = 1..7]
6.1b  PASS  complement = antipode; quotient RP^(n-1): every GF(2) Betti number 1  [n = 2..7]
6.2   PASS  lamp view: S^(n-1) at even n, contractible cone at odd n; top cells = 2^(n/2) lamp vectors  [n = 2..8]
6.3a  PASS  J, C, tau, F_ret, F_c, F_rev keep every edge; P tears exactly n2^(n/2) / (n-1)2^((n+1)/2) edges, jump n-1; T
tears  [every edge, n = 3..14 (T to 12)]
6.3b  PASS  J, C, P fix every lamp  [every word, n = 8, 10]
7.1a  PASS  Lambda_q intertwines tau, tau^-1, J, C, P  [6132 lifts (s = 2..10, q = 3,5,7) + 200 at s = 64, 128]
7.1b  PASS  canon(Lambda u) = Lambda canon u; face kept; strictly increasing  [same]
7.1c  PASS  center inherited: r_qs(Lambda u) = r_s(u), s even  [same]
7.1d  PASS  Hamming distance multiplied by q  [all pairs s <= 6; 200 wide pairs]
7.1e  PASS  even q: tau(Lambda u) = Lambda(tau u) xor 1, every u  [s = 2..8, q = 2, 4, 6]
7.2   PASS  lamps of the lift = (g g^R)^k g; route of the lift = these lamps  [4172 lifts]
7.3a  PASS  trace period p: n/p odd, x = Lambda_(n/p)(top p), orbit length 2p, census by Mobius inversion  [every word, n
= 2..16 even]
7.3b  PASS  period spectra: 192 -> {64, 192}; 576 -> {64, 192, 576}; 960 -> {64, 192, 320, 960}  [exact counts]
7.3c  PASS  running example: Lambda_3(u) has trace period 64; 63 readings + face return u  [192 bits]
7.3d  PASS  wide locks: period read by the prefix function, full return  [s = 64, 128; q up to 15 (1,920 bits)]
7.4a  PASS  canonical order kept under Lambda; partner faces mirror; gaps (137,391) (2083,7681) (32903,129013)  [s =
2..10, q = 3,5,7]
7.4b  PASS  counting address = built ring; running example at 64, 192, 320 bits = GPT values  [every word n = 2..12]
8.1   PASS  defect coordinates bijective; Delta lanes = cyclic differences, XOR 0 (q odd) / ones (q even); Delta = 0 iff
lock  [1,332,288 states exhaustive + q = 2..8 closure]
8.2a  PASS  one tick of F_c in (u, E): the skew product; inverse from the inserted pair  [same + 240 wide states]
8.2b  PASS  after s/2 ticks: E -> M_q E and u -> C(u) xor E_1  [240 wide states (s = 64, 128)]
8.2c  PASS  M_q has order exactly q  [q = 3..15 odd]
8.3   PASS  Delta(F_c X) = rotl2 Delta(X) (weight conserved); Delta(F_b X) = rotl2 Delta(X) ^ eps ^ (eps << s)
[exhaustive + wide; 5376 input checks]
8.6   PASS  local center r_qs(X) = r_s(u) xor r_s(E_k)  [every state (s even) + wide]
8.5   PASS  arrival law: center at h/2, h/2+1 (h even) or (h+1)/2 and eps=3 unseen (h odd), period h; lane 0 at (q-
1)s/2+1  [s in {2..10, 62..66, 126..130}, q = 3,5,7, 297 pulses]
8.7   PASS  majority corrects floor((q-1)/2) per column; palindromic lock: P after a fault is wrong, correct-then-P is
right  [1784 single faults, q = 5 all double faults]
8.8a  PASS  tagged shift: J(y) = rev2(b) 2^(n-2) + J(x)/4; outer two shells decide the face unless tied; x returns from
(y, a)  [every word and pair, n = 4..12]
8.8b  PASS  stream gate: 147 resolved, 69 ties (7 at 11, 16 negative jumps)  [Ovaltine 104 + Singularity 112 shifts]
8.8c  PASS  Panel C: first/last lit shell and span = GPT's (unprinted) values; uniform baseline 1/30/30  [both streams]
8.9   PASS  rank = c - c^2/2N + (pi_c - pi/2) + e, -h-1/2 < e < h/2+1/2; half-ring jump = 3/8 - c/2N + (1/4 -
A0/2^32)/2^32  [every canonical word n = 4..16; every stream word; Singularity counterfactual]
```

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github.com/QuHarmonics/The-Nexus-Harmonic-Reality info@quharmonics.com

```
51 checks, 51 passed, failed: []
ALL CHECKS PASS · seal 009815944fd8951cf95ca6bd56728f54b5e20634dd03e2d8cc67e005019d1dea
```

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github.com/QuHarmonics/The-Nexus-Harmonic-Reality info@quharmonics.com